SageMath: The Laboratory for Scientific Computing

We began this series with a simple question: why should someone already fluent in Python bother learning SageMath? Fourteen articles later, we have traversed graph theory, cybersecurity, cryptography, and the opening landscapes of number theory, and in doing so, I hope the answer has become self-evident. SageMath is not merely a mathematical dialect of Python; it is a fully equipped laboratory for scientific computing that reveals new capabilities the deeper one ventures into it. In this 15th and last article in the series, we shall explore a few more beautiful corners of that laboratory.

In the previous article of this series (published in the April 2026 issue of *OSFY*), we began our exploration of number theory and examined several number-theoretic functions provided by SageMath. Throughout this series, we have seen that prime numbers occupy a central place in computer science, particularly in algorithms related to cryptography, cybersecurity, and computational number theory. Consequently, primality testing is one of the most fundamental operations performed by mathematical software. In the previous article, we explored several functions offered by SageMath for primality testing and related computations. In this article, we will turn our attention to a few famous number-theoretic conjectures to appreciate the beauty and elegance of this branch of mathematics. Number theory is unique in that it contains centuries-old unsolved problems whose statements are so simple that they can be understood not only by professional mathematicians but even by high school students, yet they continue to challenge some of the greatest mathematical minds.

One of the oldest unsolved problems in number theory is the Goldbach Conjecture, proposed by the Prussian mathematician Christian Goldbach in 1742. It states that every even integer greater than 2 can be expressed as the sum of two prime numbers. For example, 10=3+7, 20=3+17, and 100=3+97. Although the conjecture has been verified computationally for extremely large numbers, no general mathematical proof has yet been discovered. Using SageMath, we can easily investigate the conjecture for any given even integer. The simple SageMath program *goldbach48.sage*, shown below, searches for a pair of prime numbers whose sum equals the given even integer and returns the first such pair that it finds. Notice that the program relies on just two built-in SageMath functions, *prime_range( )* and *is_prime( )*. The function *prime_range( )* efficiently generates prime numbers within a specified range, while *is_prime( )* determines whether a given integer is prime. Both functions are highly optimized in SageMath, allowing the program to test the conjecture for very large integers much more efficiently than a straightforward implementation using basic Python code.